

Store procedure dengan output parameter

ISA212 Basisdata 2
minggu 6

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Contoh tabel

Id	Name	Gender	DepartmentId
1	Smith	Male	1
2	Ram	Male	1
3	Kevin	Female	3
4	Jack	Male	2
5	Smith	Male	3
6	Olivia	Female	2
7	Jacob	Male	1
8	Neymar	Male	2
9	Muller	Male	1
10	Priyanka	Female	2

Employee table

```
create database ISA212_minggu6
```

```
go
```

```
use ISA212_minggu6
```

```
go
```

```
create table employee
```

```
(ID int primary key , name varchar(40) , Gender varchar(20) ,  
departement_id int)
```

Store procedure with output parameter

To create a stored procedure with output parameter, we use the keywords OUT or OUTPUT. @EmployeeCount in the below stored procedure is an OUTPUT parameter. Notice that it is specified with OUTPUT keyword.

```
CREATE PROCEDURE spGetEmployeeCountByGender
@Gender nvarchar(20),
@EmployeeCount int Output
AS
BEGIN
SELECT @EmployeeCount = COUNT(Id)
FROM tblEmployee
WHERE Gender = @Gender
END
```

Store procedure with output parameter

To execute this stored procedure with OUTPUT parameter, follow the below steps:

1. First, initialise a variable of same datatype as that of the output parameter. Here, we have declared @EmployeeTotal integer variable.
2. Then pass the @EmployeeTotal variable to the stored procedure. You have to specify the OUTPUT keyword. If you don't specify the OUTPUT keyword, the variable will be NULL.
3. Then execute the stored procedure.

Execute Store procedure with output parameter

```
DECLARE @EmployeeTotal int
EXECUTE spGetEmployeeCountByGender 'Female',
@EmployeeTotal output
PRINT @EmployeeTotal
```

If you don't specify the OUTPUT keyword, while executing the stored procedure, the @EmployeeTotal variable will be NULL. In the example below, we have not specified OUTPUT keyword. So while executing the stored procedure, a message of @EmployeeTotal is null is printed.

Execute Store procedure with output parameter

```
DECLARE @EmployeeTotal int
EXECUTE spGetEmployeeCountByGender 'Female', @EmployeeTotal

IF(@EmployeeTotal is null)
PRINT '@EmployeeTotal is null'
ELSE
PRINT '@EmployeeTotal is not null'
```


Execute Store procedure with output parameter

While using the parameter names, you can pass the parameters in any order. In the example below, we are first passing the OUTPUT parameter and then the input @Gender parameter. But we will get the total number of male employees without any errors.

```
DECLARE @EmployeeTotal int
EXECUTE spGetEmployeeCountByGender @EmployeeCount =
@EmployeeTotal OUT, @Gender = 'Male'
PRINT @EmployeeTotal
```


Execute Store procedure with output parameter

- Now let's have a quick look at some of the extremely useful system stored procedures.
- `sp_help SP_Name`: Used to view the information about the stored procedure like parameter names, their datatypes, etc. `sp_help` can be used with any database object, like Tables, Views, SPs, Triggers, etc. Alternatively, you can also press ALT+F1, when the name of the object is highlighted.

Sp_HELP <sp_helpname>

SQLQuery1.sql - (L...ampleDB (sa (52)))

```
sp_help spGetEmployeeCountByGender
```

Results Messages

	Name	Owner	Type	Created_datetime
1	spGetEmployeeCountByGender	dbo	stored procedure	2014-07-06 20:46:15.247

	Parameter_name	Type	Length	Prec	Scale	Param_order	Collation
1	@Gender	nvarchar	40	20	NULL	1	SQL_Latin1_General_CP1_CI_AS
2	@EmployeeCount	int	4	10	0	2	NULL

Sp_help <table_name>

- You can use sp_help with any database objects like Tables, Views, Triggers, etc. For example, when we use sp_help with tblEmployee table, we will get all the information about the table like different columns present in the table, their data types, indexes associated with the table, constraints associated with the table, etc.

SQLQuery1.sql - (L...ampleDB (sa (52)))

```
sp_help tblEmployee
```

Results Messages

	Name	Owner	Type	Created_datetime
1	tblEmployee	dbo	usertable	2014-01-19 22:49:54.943

	Column_name	Type	Computed	Length	Prec	Scale	Nullable	TrimTrailingBlanks	FixedLenNullInSource	Collation
1	Id	int	no	4	10	0	no	(n/a)	(n/a)	NULL
2	Name	nvarchar	no	100			yes	(n/a)	(n/a)	SQL_Latin1_General_CP1_CI_AS
3	Gender	nvarchar	no	100			yes	(n/a)	(n/a)	SQL_Latin1_General_CP1_CI_AS
4	DepartmentId	int	no	4	10	0	no	(n/a)	(n/a)	NULL

	Identity	Seed	Increment	Not For Replication
1	No identity column defined.	NULL	NULL	NULL

	RowGuidCol
1	No rowguidcol column defined.

	Data_located_on_Negroup
1	PRIMARY

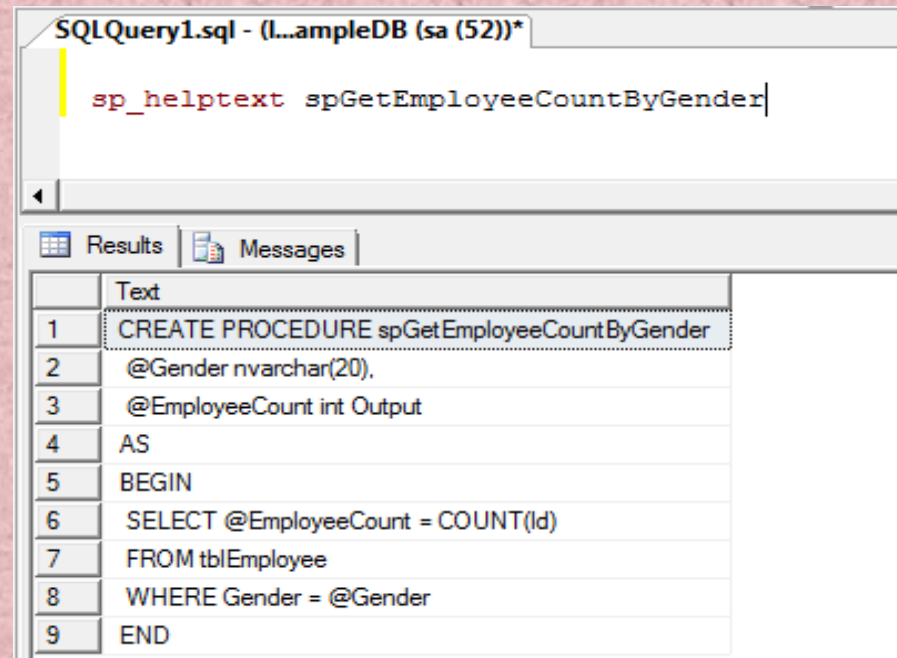
	index_name	index_description	index_keys
1	PK_Employee	clustered, unique, primary key located on PRIMARY	Id

	constraint_type	constraint_name	delete_action	update_action	status_enabled	status_for_replication	constraint_keys
1	PRIMARY KEY (clustered)	PK_Employee	(n/a)	(n/a)	(n/a)	(n/a)	Id

Sp_helpText

sp_helptext SP_Name: Used to view the Text of the stored procedure.

For example, when we use `sp_helptext spGetEmployeeCountByGender`, we will get the text of this stored procedure.

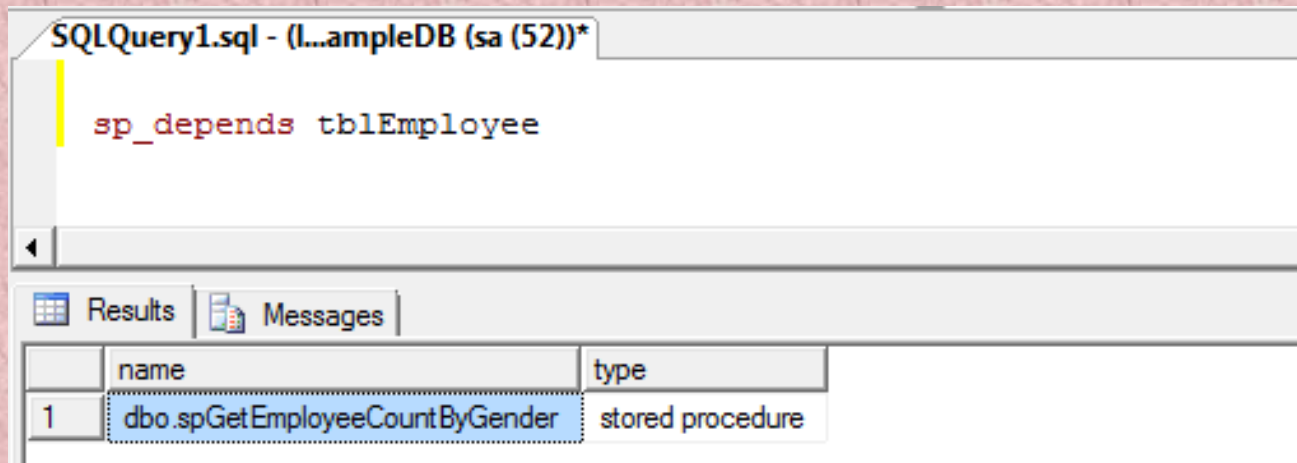


The screenshot shows a SQL Server Enterprise Manager window titled "SQLQuery1.sql - (L...ampleDB (sa (52)))". The query editor contains the command `sp_helptext spGetEmployeeCountByGender`. Below the editor, the "Results" tab is active, displaying the text of the stored procedure in a table with two columns: "Text" and "Line". The text is as follows:

Line	Text
1	CREATE PROCEDURE spGetEmployeeCountByGender
2	@Gender nvarchar(20),
3	@EmployeeCount int Output
4	AS
5	BEGIN
6	SELECT @EmployeeCount = COUNT(Id)
7	FROM tblEmployee
8	WHERE Gender = @Gender
9	END

SP_DEPENDS

`sp_depends SP_Name`: Used to view the dependencies of the stored procedure. This system stored procedure is very useful, especially if you want to check whether there are any stored procedures that are referencing a table which you are about to drop. `sp_depends` can also be used with other database objects like Tables, Views, etc.



The screenshot shows a SQL query window titled "SQLQuery1.sql - (L...ampleDB (sa (52)))". The query entered is `sp_depends tblEmployee`. Below the query window, the "Results" tab is active, displaying a table with two columns: "name" and "type". The table contains one row with the values "dbo.spGetEmployeeCountByGender" and "stored procedure".

	name	type
1	dbo.spGetEmployeeCountByGender	stored procedure

